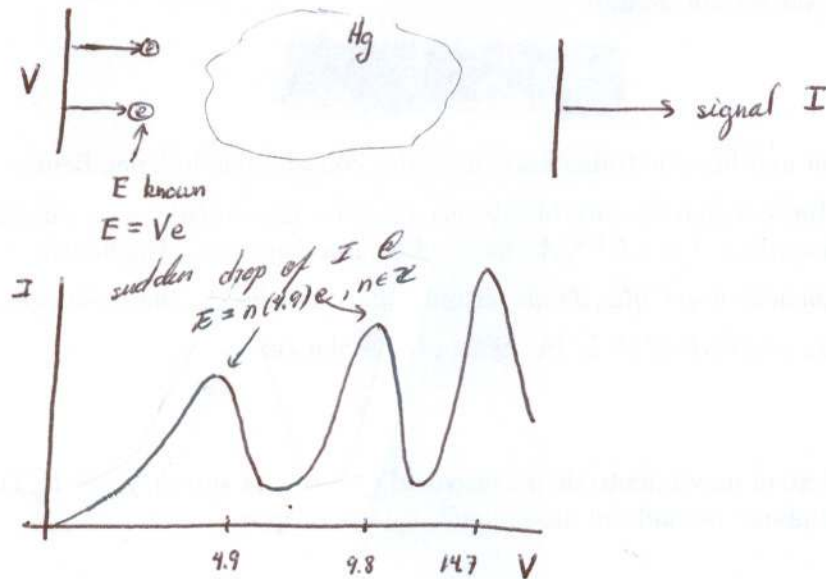


~ Franck-Hertz experiment

The success of Bohr's model was sealed by this experiment (1914) where Franck-Hertz demonstrated that energy-levels do exist. Basically, they confirmed that elements absorb precise energies.



The sudden fall of I @ $\bar{E} = n(4.9)e$ was interpreted as the absorption of T_e by the Hg-atoms, whose ionization energy is precisely 4.9 eV!

~ Bohr's puzzles

Even though Bohr's model is quite successful, some questions remained a challenge:

- Why does it only work in a precise manner for H-like atoms? (It fails for He, e.g.!)
 - Why are some spectral lines more intense?
 - Why do some lines consist of several smaller lines more detectable under magnetic fields?
 - Why are emissions related to brighter spectral lines more frequent?
- At a more fundamental/theoretical level, where ^{do} such quantizations as $r_n = a_0 n^2$ come from?

→ Quantum mechanics.

In the Copenhagen interpretation, quantum mechanics mathematical formulation is based in the following principles

1. A system is completely described by a wave function ψ
2. How ψ changes (in space and time) is described by the Schrödinger equation
3. The quantum description of nature is essentially probabilistic and the Born rule: probability^(density) of an event (at a point in space & an instant) is given by $P = |\psi|^2 \rightarrow \psi$ is the probability amplitude
4. Heisenberg's uncertainty principle. It's impossible to know the values of all properties of the system simultaneously.
5. Matter exhibits a wave-particle duality.
6. Correspondence principle.
- 7* It's impossible to determine the exact "state" of ψ before any measurement has taken place. Principle of wave-function collapse. Schrödinger's cat.
- 8* Principle of spectral decomposition

* This are not really part of the Copenhagen interpretation

~ Wave function in Hilbert spaces

A Hilbert space is a finite or infinite dimensional ^{vector} space endowed with an inner product, satisfying the basic rules:

1. ^{Conjugate} Symmetric in its both arguments
2. ^(alt) Linear in the first argument and linear in the second one if they are (complex) real.
3. Positive definite

The inner product associates a real or complex number to two elements of the Hilbert space.

A typical example of Hilbert spaces is the Euclidean space, whose elements are ~~the~~ vectors $\{|r\rangle\}$

A more interesting example of Hilbert space is the space of wave-functions $\{|\psi\rangle\}$ ^{ket}

The inner product is denoted by bra $\langle\varphi|\psi\rangle$ ^{ket}

which may have at least ~~two~~ ^{two} interpretations:

- a) the projection of $|\psi\rangle$ along $|\varphi\rangle$
- b) the representation of $|\psi\rangle$ in the "direction" $|\varphi\rangle$

→ DISCUSSION IN (2d)

If the space of $\{|\psi\rangle\}$ is infinite dimensional & continuous, the inner product allows one to perform a change of basis as

$$|\psi\rangle \equiv \int d^D\varphi \overset{\text{dimension of element } |\varphi\rangle}{|\varphi\rangle \langle\varphi|\psi\rangle}$$

provided that $\{|\varphi\rangle\}$ is a complete ^{continuous} basis, (and continuous), i.e. satisfying the closure relation

$$\int d\varphi |\varphi\rangle \langle\varphi| = 1 \quad \left\{ \begin{array}{l} \text{identity operator} \\ \text{Completeness or closure relation} \end{array} \right.$$

and orthonormal, i.e.

$$\langle\varphi'|\varphi\rangle = \delta(\varphi - \varphi')$$

The basis $\{|\varphi\rangle\}$ is acceptable as long as any $|\psi\rangle$ has a unique expression in terms of $\{|\varphi\rangle\}$.

$|\psi\rangle$ can also be expressed in terms of a discrete basis $\{|s_i\rangle\}$, $i \in \mathbb{Z}$

$$|\psi\rangle \equiv \sum_i |s_i\rangle \langle s_i|\psi\rangle$$

as long as

$$\sum_i |s_i\rangle \langle s_i| = 1 \quad \text{Completeness}$$

$$\langle s_i | s_j \rangle = \delta_{ij} \quad \text{Orthonormality}$$

(24)

A useful basis is $\{|\vec{r}\rangle\}$. Let us define the projection of $|\psi\rangle$ on $|\vec{r}\rangle$ as

$$\psi(\vec{r}) \equiv \langle \vec{r} | \psi \rangle$$

$$\Rightarrow |\psi\rangle = \int d^3r \psi(\vec{r}) |\vec{r}\rangle$$

(23) ← In Dirac's notation, the inner product must satisfy

$$\langle \varphi | \psi \rangle = \langle \psi | \varphi \rangle^*$$

$$\langle \varphi | \lambda_1 \psi_1 + \lambda_2 \psi_2 \rangle = \lambda_1 \langle \varphi | \psi_1 \rangle + \lambda_2 \langle \varphi | \psi_2 \rangle$$

$$\langle \lambda_1 \varphi_1 + \lambda_2 \varphi_2 | \psi \rangle = \lambda_1^* \langle \varphi_1 | \psi \rangle + \lambda_2^* \langle \varphi_2 | \psi \rangle$$

$$\langle \psi | \psi \rangle \text{ real positive, } \langle \psi | \psi \rangle = 0 \Leftrightarrow |\psi\rangle = 0$$

Notice that, using the completeness relations & properties of inner product

$$\langle \psi | \psi \rangle = \int d^3r \langle \psi | \vec{r} \rangle \langle \vec{r} | \psi \rangle = \int d^3r \psi^*(\vec{r}) \psi(\vec{r}) \geq 0$$

We notice that $\langle \psi | \psi \rangle$ is satisfied because

$$|\psi(\vec{r})|^2 \geq 0$$

Since $P = |\psi|^2$, then ψ must be normalized

$$\int d^3r P = \int d^3r |\psi|^2 = 1$$

which, in this basis, means that the system is definitely somewhere.

Further, the probability that the system described by $\psi(\vec{r})$ be between $\vec{r}_1$ & $\vec{r}_2$ is

$$\int_{\vec{r}_1}^{\vec{r}_2} |\psi|^2$$

Since $\psi(\vec{r})$ is normalizable, then

$$\lim_{r \rightarrow \infty} \psi(\vec{r}) = 0$$

Additional considerations constrain ψ to fulfill following properties:

- ψ must be continuous & single-valued everywhere (follow from basis constr.)
- Partial derivatives of ψ must also be single-valued & continuous (This condition is not always satisfied!)

~ Schrödinger's equation

Intrinsically, $|\psi\rangle$ can have a time-dependence, such that

$$|\psi\rangle = |\psi(t)\rangle.$$

Thus, so do ψ

$$\psi(\vec{r}, t) = \langle \vec{r} | \psi(t) \rangle$$

Schrödinger's equation governs the time-evolution of $|\psi\rangle$ (and thus also of $\psi(\vec{r}, t)$). In the basis of $\{|\vec{r}\rangle\}$, the resulting wave function turns out to be a wave ~~equation~~. Recalling that the ^(a)solution to

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$$

moving along the $+x$ direction is

$$y(x, t) = A e^{\text{complex amplitude}},$$

it's tempting to propose for a free particle

$$\psi(x, t) = A e^{-i\omega(t-x/v)}$$

with $v \equiv \lambda \nu$. Using $(\hbar \equiv h/2\pi)$

$$E = h\nu = 2\pi \hbar \nu \quad \text{and} \quad \lambda = \frac{h}{p} = \frac{2\pi \hbar}{p}$$

$$\Rightarrow \psi(x, t) = A e^{-2\pi i(\nu t - x/\lambda)} = A e^{-\frac{i}{\hbar}(Et - px)}$$

$$\text{or analogously} \quad \psi(\vec{r}, t) = A e^{-\frac{i}{\hbar}(Et - \vec{p} \cdot \vec{r})}$$

From our proposed solution, we observe that

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{p^2}{\hbar^2} \psi \quad \& \quad \frac{\partial \psi}{\partial t} = -\frac{iE}{\hbar} \psi$$

$$\Leftrightarrow p^2 \psi = -\hbar^2 \frac{\partial^2 \psi}{\partial x^2} \quad \& \quad E \psi = i\hbar \frac{\partial \psi}{\partial t}$$

or $\hat{p}^2 \psi = -\hbar^2 \nabla^2 \psi \quad \& \quad E \psi = i\hbar \frac{\partial \psi}{\partial t}$

On the other hand, we know that the total energy of a free particle is given by

$$E = \frac{p^2}{2m}$$

which, multiplying by ψ , leads to

$$E \psi = \frac{p^2}{2m} \psi$$

$\Leftrightarrow$

$$\boxed{i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi}$$

If we generalize this result to interacting systems with potential energy $V(\vec{r})$, we obtain

$$E \psi = \left(\frac{p^2}{2m} + V \right) \psi$$

Hamiltonian

$$\Leftrightarrow \boxed{i\hbar \frac{\partial \psi}{\partial t} = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi}$$

This is, of course, not a derivation of the Schrödinger eq. In fact, there is no possible derivation, as it is a fundamental equation, ⁱⁿ as much as the second Newton's law. Notice that $p\psi = -i\hbar \frac{\partial \psi}{\partial x}$ (or $\hat{p}\psi = -i\hbar \vec{\nabla}\psi$).

- Schrödinger equation in Dirac's notation

It's clear that the relations

$$p^2 \psi = -\hbar^2 \frac{\partial^2 \psi}{\partial x^2}, \quad p\psi = -i\hbar \frac{\partial \psi}{\partial x}$$

only make sense for a solution $\psi = \psi(x) = \langle x | \psi \rangle$.

BUT this is only a representation in the space $\{|x\rangle\}$ of $|\psi\rangle$. Thus, one

is compelled to notice that

$$p^2 \equiv -\hbar^2 \frac{\partial^2}{\partial x^2} \quad (\text{or } p^2 \equiv -\hbar^2 \nabla^2) \quad \& \quad p \equiv -i\hbar \frac{\partial}{\partial x} \quad (\text{or } \vec{p} \equiv -i\hbar \vec{\nabla})$$

can be considered the differential representation in the space $\{|x\rangle\}$ (or $\{|\vec{r}\rangle\}$) of

an operator p^2 . Meaningful operators on a vector space must be linear, such that, if $\hat{A}$ is a linear operator on $\{|\psi\rangle\}$, then

$$|\psi'\rangle = \hat{A}|\psi\rangle \quad |\psi'\rangle \in \{|\psi\rangle\}$$

$$\hat{A}(\lambda_1|\psi_1\rangle + \lambda_2|\psi_2\rangle) = \lambda_1 \hat{A}|\psi_1\rangle + \lambda_2 \hat{A}|\psi_2\rangle$$

In these terms, it seems natural that the abstract form of p^2 acting on $|\psi\rangle$ together with a corresponding form of E & V be linear operators, satisfying

$$\boxed{\hat{E}|\psi\rangle = \hat{H}|\psi\rangle \equiv \left(\frac{\hat{p}^2}{2m} + \hat{V} \right) |\psi\rangle}$$

Multiplying by $\langle \vec{r} |$ on the left leads to the representation of the Schrödinger eq. in $|\vec{r}\rangle$ -space:

$$\langle \vec{r} | \hat{E} | \psi \rangle = i\hbar \frac{\partial \psi(\vec{r}, t)}{\partial t}$$

$$= \langle \vec{r} | \left(\frac{\hat{p}^2}{2m} + \hat{V} \right) | \psi \rangle = \left(\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \right) \psi(\vec{r}, t)$$

Note to
- Other representations.

As we mentioned, the basis $\{|\vec{r}\rangle\}$ is a useful one, but not the only one. Another useful basis is $\{|\vec{p}\rangle\}$ on which

$$\langle \vec{p} | \psi \rangle = \psi(\vec{p}) \quad \& \quad i\hbar \frac{\partial \psi(\vec{p}, t)}{\partial t} = \left(\frac{p^2}{2m} + V(\vec{p}) \right) \psi(\vec{p}, t)$$

The precise form of $\psi(\vec{p}, t)$, $V(\vec{p})$ as well as their relations to $\psi(\vec{r}, t)$ & $V(\vec{r})$ will be studied later

- The stationary Schrödinger equation. (time-independent)

If the energy of a system does not change in time, i.e.

$$\frac{d\hat{V}}{dt} = 0$$

$$\Rightarrow \boxed{E|\psi\rangle = \left(\frac{\hat{p}^2}{2m} + \hat{V}\right)|\psi\rangle}$$

$$\text{or } E\psi(\vec{r}, t) = \left(-\frac{\hbar^2}{2m}\nabla^2 + V(\vec{r})\right)\psi(\vec{r}, t)$$

→ see (29)

- Superposition.

Since $\hat{E}$, $\hat{V}$, $\hat{p}^2$ are linear operators, it's clear that Schrödinger's equation is linear ^{in $|\psi\rangle$} too (it's actually evident from the form). Thus, if ψ_1 & ψ_2 are two solutions, then

$$\psi = \lambda_1 \psi_1 + \lambda_2 \psi_2 \quad \lambda_1, \lambda_2 \in \mathbb{C}$$

is a solution too.

This implies that the wave functions will lead to probability-interference terms because

$$P = |\psi|^2 = |\lambda_1 \psi_1 + \lambda_2 \psi_2|^2 = \lambda_1^2 |\psi_1|^2 + \lambda_2^2 |\psi_2|^2 + (\lambda_1 \lambda_2^* \psi_1 \psi_2^* + \text{h.c.})$$

This offers immediately an explanation to the result of the double-slit experiment.



(28) ← Consider ^{e.g.} the wavefunction for a free particle in one dim. with $V = V(x)$
 $\psi(x, t) = \psi_x(x) e^{-\frac{i}{\hbar} E t}$ $\psi_x(x) \equiv A e^{+\frac{i}{\hbar} p x}$

$$\Rightarrow i\hbar \frac{\partial \psi}{\partial t} = E \psi_x(x) e^{-\frac{i}{\hbar} E t}$$

$$= e^{-\frac{i}{\hbar} E t} \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \psi_x(x)$$

$$\Leftrightarrow E \psi_x = \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right) \psi_x$$

- Expectation values

The expectation value of a particular variable A of a quantum system is denoted by

$$\langle \hat{A} \rangle$$

What does it mean? The ~~most likely~~ ^{average} value of A in the system ^{among a discrete set of pos.}
 Consider as a warm-up example that we want to figure out the ~~most~~ ^{average} ~~likely~~ place where we can find a particle provided a system of a large number ^{M} of identical particles. If N_i denotes the number of particles at the position x_i & $M = \sum_i N_i$, then the weighted average of positions x_i

$$\bar{x} = \frac{N_1 x_1 + N_2 x_2 + \dots}{M} = P_1 x_1 + P_2 x_2 + \dots$$

as the most likely value of x where to find a particle. In this expression, we have defined $P_i \equiv \frac{N_i}{M}$ as the probability of finding a particle at x_i .

If $\{x_i\}$ was a continuous set of positions, we'd have instead

$$\bar{x} = \int_{-\infty}^{\infty} P(x) x dx$$

Since in quantum systems

(20)

$$P(x) = |\psi(x)|^2$$

then we expect e.g.

$$\langle x \rangle = \int dx \psi^*(x) x \psi(x)$$

or in general

$$\langle \vec{r} \rangle = \int d^3r \psi^*(r) r \psi(r)$$

Relevant literature:

pages 149, 150 of
Cohen-Tannoudji

How can we generalize this approach to any other expectation value. It will be most convenient to use Dirac notation. In this notation, given a system described by $|\psi\rangle$, the expectation value of $\hat{A}$ (as operator) is

$$\begin{aligned} \langle \hat{A} \rangle &= \langle \psi | \hat{A} | \psi \rangle = \langle \psi | \mathbb{1} \hat{A} | \psi \rangle \\ &= \int d^3r \langle \psi | r \rangle \langle r | \hat{A} | \psi \rangle \end{aligned}$$

Let's define

$$\langle r | \hat{A} | \psi \rangle \equiv A(r) \langle r | \psi \rangle = A(r) \psi(r)$$

↖ eigenvalue of $\hat{A}$

where $A(r)$ is the eigenvalue (function) of the abstract operator $\hat{A}$ in the representation of $\{|r\rangle\}$. Then

$$\langle \hat{A} \rangle = \int d^3r \psi(r)^* A(r) \psi(r)$$

→ (32)
OBS.

$$\langle \hat{A} \rangle = \langle \psi | \hat{A} | \psi \rangle = \int d^3r \langle \psi | \hat{A} | r \rangle \langle r | \psi \rangle$$

$$\langle \psi | \hat{A} | r \rangle \stackrel{?}{=} (\langle r | \hat{A} | \psi \rangle)^* = (A(r) \psi(r))^* = \psi^*(r) A^*(r)$$

$$\Rightarrow \langle \hat{A} \rangle = \int d^3r \psi^*(r) A^*(r) \psi(r) \stackrel{!}{=} \int d^3r \psi^*(r) A(r) \psi(r)$$

Thus, we have

$$A^*(r) = A(r)$$

meaning that $A(r)$ must be real. I.e. the eigenvalues of an operator $\hat{A}$ in a given representation must be real in order to be admissible as an operator whose expectation value can be computed!

This implies that the associated operators must be real too:

$$\langle \hat{A} \rangle \stackrel{!}{=} \langle \hat{A} \rangle^* \Rightarrow \left(\langle \psi | \hat{A} | \psi \rangle \right)^* = \langle \psi | \hat{A}^\dagger | \psi \rangle = \langle \psi | \hat{A} | \psi \rangle \\ \stackrel{!}{=} \langle \psi | \hat{A} | \psi \rangle$$

$$\Leftrightarrow \hat{A} = \hat{A}^\dagger \quad \text{Hermitian operator}$$

In other words, Hermitian operators have real eigenvalues, and these eigenvalues can be measured! $\leadsto$ describe observables!

Obviously, the operators $\hat{x}$, such that

$$\langle x | \hat{x} | \psi \rangle = x \psi(x),$$

is Hermitian and has eigenvalues that are real. Can we say the same about the operator $\hat{p}_x$? (See homework - page 151 Cohen-Tannoudji)

The first observation on this is that order matters. In the position-representation, we have

$$\langle \hat{p}_x \rangle = \int d^3r \psi^*(r) (-i\hbar) \frac{\partial}{\partial x} \psi(r)$$

where we have used

$$\langle r | \hat{p}_x | \psi \rangle = -i\hbar \frac{\partial}{\partial x} \psi(r) \equiv p_x(r) \psi(r)$$

Clearly, the representation of $\hat{p}_x$ in $\{|r\rangle\}$ cannot be located anywhere in the integral. E.g.

$$-i\hbar \int d^3r \frac{\partial}{\partial x} (\psi^*(r) \psi(r)) = -i\hbar \int dy dz \int_{-\infty}^{\infty} dx \frac{\partial}{\partial x} (\psi^*(r) \psi(r)) = 0$$

which makes no sense! (We used that $\lim_{x \rightarrow \pm\infty} \psi(x) = 0$)

The second observation is that e.g. in the free particle

$$\psi(x,t) = e^{-\frac{i}{\hbar}(Et - px)}$$

$$-i\hbar \frac{\partial}{\partial x} \psi(x,t) = -i\hbar \left(+\frac{i}{\hbar}\right) p \psi(x,t) = p \psi(x,t), \quad p \in \mathbb{R}$$

However, this could be considered a circular argument (but it's NOT).
The definitive answer is encoded in the integral

$$\langle \varphi | \hat{p}_x | \psi \rangle = \int d^3r \varphi^*(r) (-i\hbar) \frac{\partial}{\partial x} \psi(r)$$

which turns out to coincide with (HOMEWORK)

$$\langle \psi | \hat{p}_x | \varphi \rangle^* = \langle \varphi | \hat{p}_x^\dagger | \psi \rangle$$

(30)

← Notation and properties of linear operators

1. $A\psi \equiv \lambda \psi$

$$\Rightarrow \langle \varphi | \lambda \psi \rangle = \lambda \langle \varphi | \psi \rangle$$

$$\Rightarrow \langle \lambda \psi | \varphi \rangle = \langle \varphi | \lambda \psi \rangle^* = \lambda^* \langle \varphi | \psi \rangle^* = \lambda^* \langle \psi | \varphi \rangle$$

$$\Leftrightarrow \boxed{\langle \lambda \psi | \quad \lambda^* \langle \psi |}$$

2.

$$\hat{A}\psi \equiv \hat{A}\psi$$

3. Adjoint of $\hat{A}$ or hermitian conjugate of $\hat{A}$

$$\langle \varphi | \hat{A} \psi \rangle = \langle \varphi | \hat{A} \psi \rangle = \int d^3r \varphi^*(r) (A\psi)(r) = \int d^3r \varphi^*(r) A(r) \psi(r) \stackrel{\text{define}}{=} \int d^3r (\varphi A^\dagger)(r) \psi(r) = \langle \hat{A}^\dagger \varphi | \psi \rangle$$

$$\boxed{\langle \varphi | \hat{A} \psi \rangle \equiv \langle \hat{A}^\dagger \varphi | \psi \rangle}$$

4.

$$\langle \varphi | \hat{A} \psi \rangle^* = \langle \varphi | \hat{A} \psi \rangle^* = \langle \hat{A}^\dagger \varphi | \psi \rangle^* = \langle \psi | \hat{A}^\dagger \varphi \rangle = \langle \psi | \hat{A}^\dagger | \varphi \rangle$$

$$\Leftrightarrow \boxed{\langle \varphi | \hat{A} | \psi \rangle = \langle \psi | \hat{A}^\dagger | \varphi \rangle}$$